

level 2

no. of sessions: 4

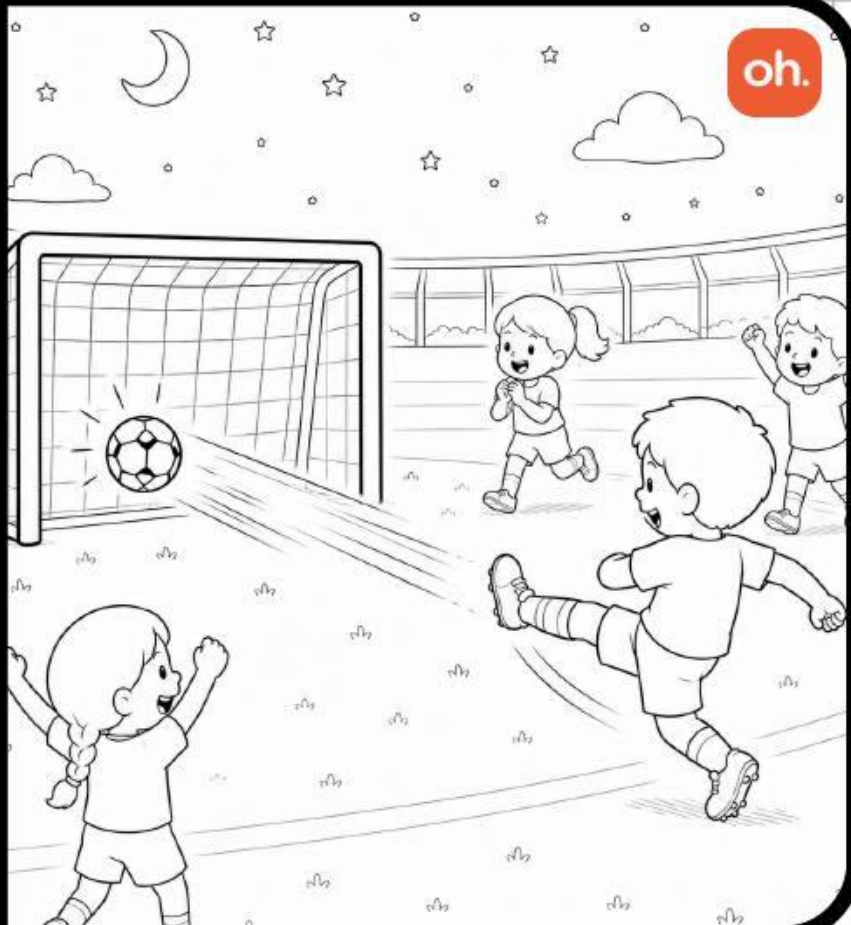
soccerbot

The soccer bot moves and pushes a ball to score goals, just like a robot playing a game of football.



simple machine: levers

electronic parts: polarity reversal





Motor Driver

The **brain of a robot**. I listen to input devices like sensors and tell output devices like motors/LEDs when to turn ON/OFF.



IR Sensor

I am like a **robot's eyes**. I help robots see objects or follow a path without touching anything.



LDR Sensor

I am like a **light detector** who tells a robot if it is bright or dark and i am used in automatic lights.



Motor

I **change electricity into movement** and help wheels, fans, and robots spin and move.



DPDT Switch Controller

Like a **direction button** - i make a motor spin forward or backward without using coding.



Servo Motor

I am a **smart motor** that can stop at the exact position you want. I am used in robot arms, gates, and moving parts.



Servo Controller

I am like a **control room** for the servo motor and lets you move the servo easily to test how it works.



Battery & Battery Holder

I **give power** to the project, and my battery holder keeps the battery safe and connected. Together, we help the model work.



Jumper Wires

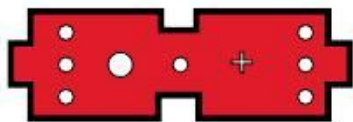
We are the **roads for electricity** who connect the different parts of the circuit, so power can travel.



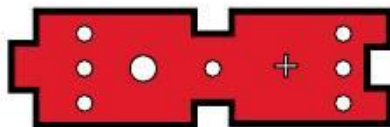
Motor Clamp

I am like a **strong hand** that holds the motor tightly in place while it spins.

materials needed



M-M beam - x1



M-F beam - x1



I clamp - x1



motor holder - x2

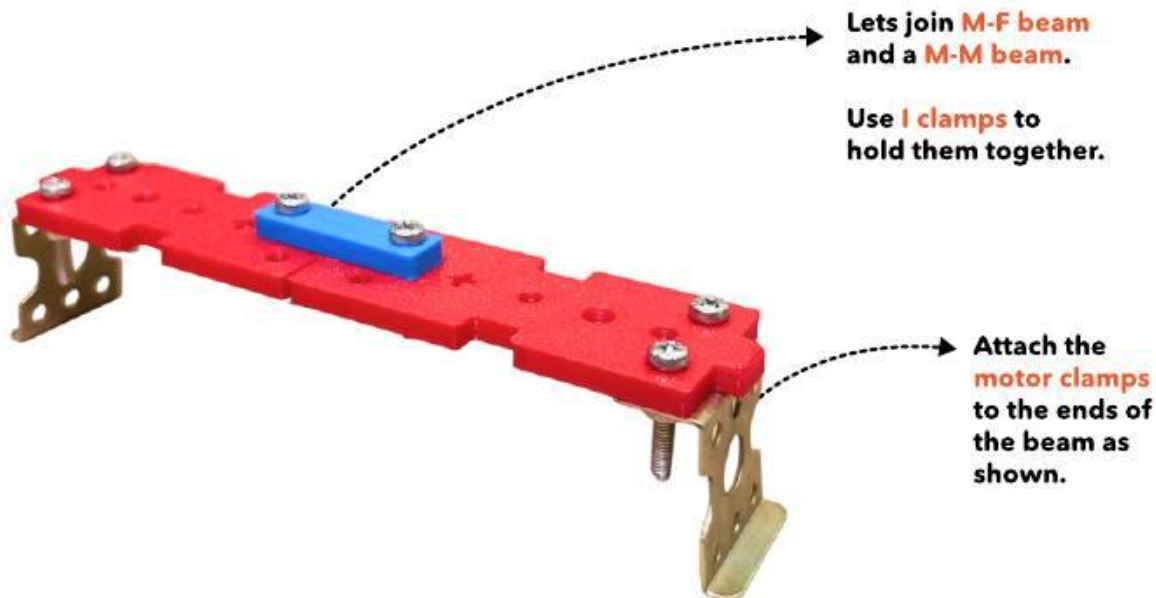


small screw - x6



nut - x6

step 1 : assemble the chassis



materials needed



motor - x2

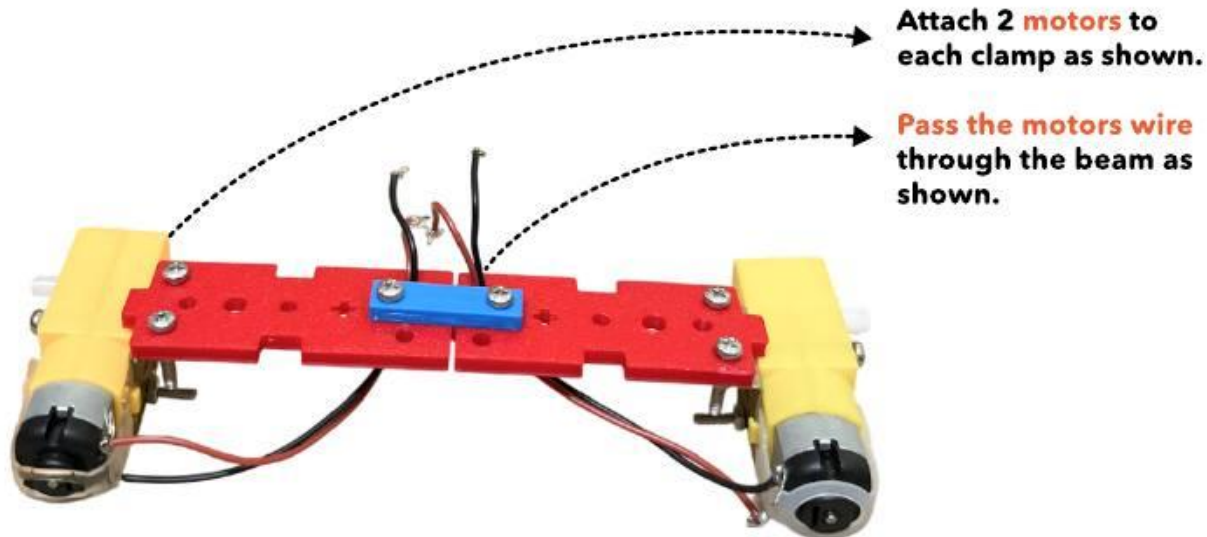


long screw - x4



nut - x4

step 2: attach the motors



materials needed



F-F beam - x2

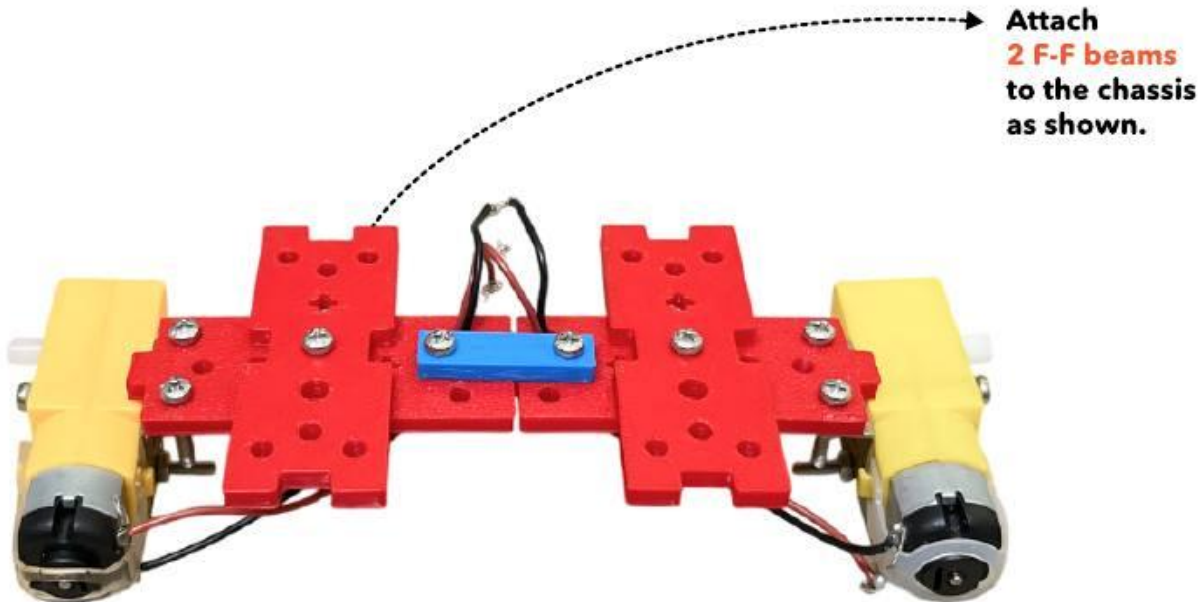


small screw - x2

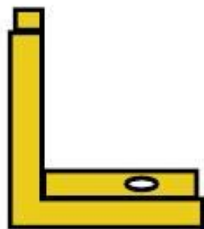


nut - x2

step 3: complete the chassis assembly



materials needed



L clamp - x2



small screw - x2



nut - x2

step 4: assemble the front supports



Add **2 L Clamps**
to the **FF beam**
as shown.

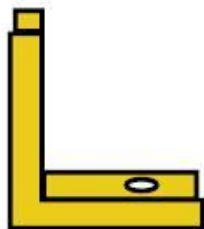
materials needed



F-F beam - x4



M-F beam - x4



L clamp - x2

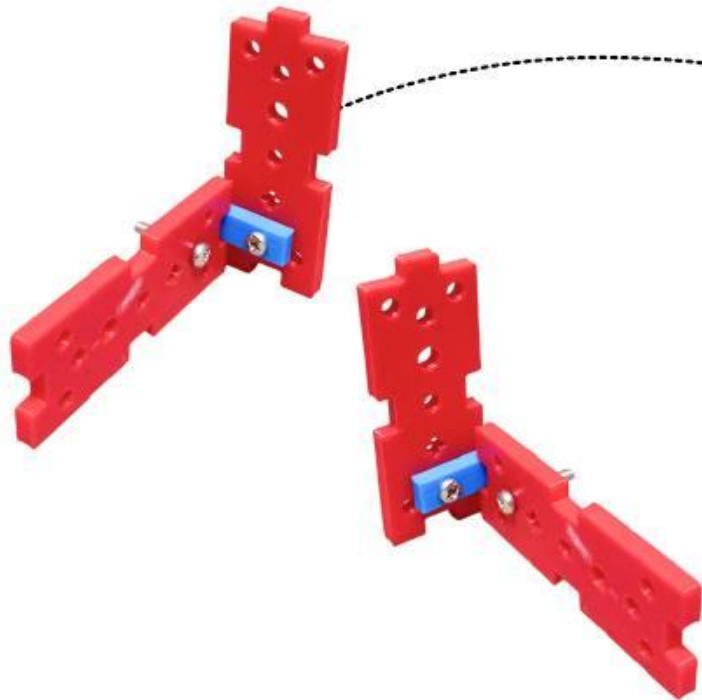


small screw - x4



nut - x4

step 5: assemble two ball holders



Take a **M-F beam** and place it vertically and a **F-F beam** placed horizontal.

Use a **L clamp** to join them together as shown.

materials needed

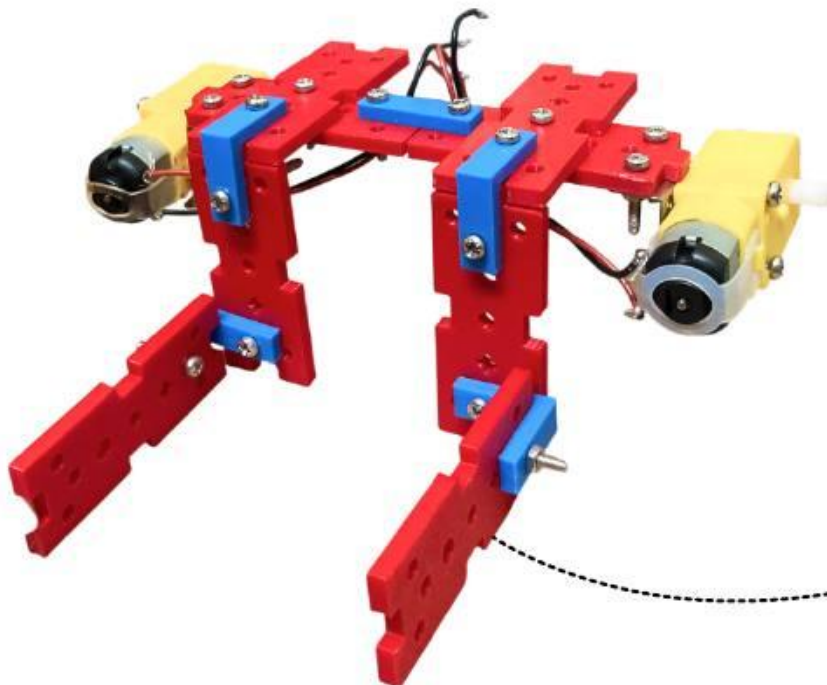


small screw - x2



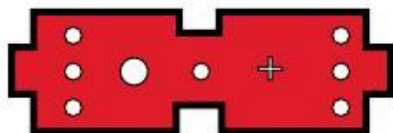
nut - x2

step 6: fix the ball holder to the chassis



Fix the two ball holder to the front as shown.

materials needed



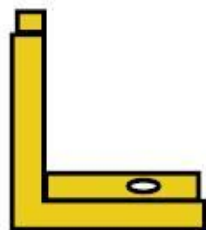
M-M beam - x2



small screw - x6



nut - x6

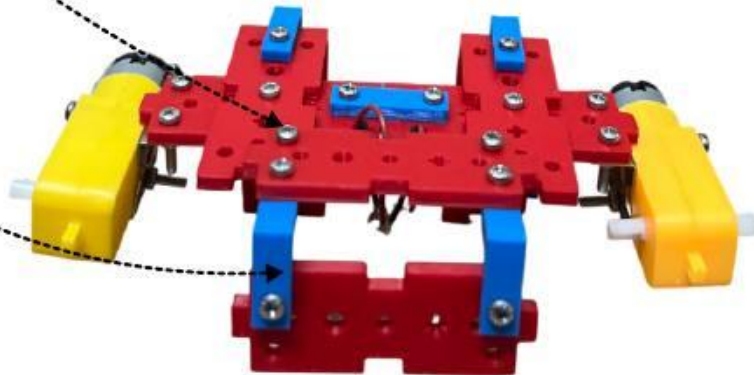


L clamp - x2

step 7: assemble the back supports

Add a **M-M beams**
using small screws

Add another
M-M beam
using a
L clamp



materials needed



wheel - x2

step 8: add the wheels



materials needed



motor controller - x2



battery holder x1



battery x1

step 9: circuit

